Selective State-Space Modeling for Joint Channel Estimation in OFDM-Based Integrated Sensing and Communication: A Mamba-ISAC Framework

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Abstract*—Integrated sensing and communication (ISAC) is a foundational pillar of sixth-generation (6G) networks, unifying radar-class environmental sensing with high-throughput data links on a shared waveform. A central bottleneck in ISAC systems is joint channel estimation: the communication channel and the sensing (target) channel are physically coupled and must be tracked jointly and rapidly under user and target mobility. Existing solutions rely on classical linear estimators (LMMSE), which do not exploit temporal structure, or on Transformer-based deep networks, whose quadratic self-attention complexity limits scalability to long pilot sequences and real-time deployment. Selective state-space models (SSMs), and Mamba in particular, offer linear-time sequence modeling with input-dependent state transitions. This paper presents Mamba-ISAC, a selective state-space architecture for joint communication-channel ($\mathbf{H}_c$) and target-parameter (range R, Doppler shift ν_s) estimation in monostatic OFDM ISAC systems. We benchmark Mamba-ISAC against closed-form LMMSE and a parameter-matched Transformer (3.45% parameter discrepancy, ~ 1.02M parameters) under a synthetic time-correlated Rician channel generator across SNR, sequence length, mobility, and pilot density sweeps. Empirical results demonstrate that Mamba-ISAC achieves target range estimation accuracy of 65.11 m RMSE and communication channel NMSE of 0.38 dB, while preserving linear $O(T)$ sequence length complexity compared to quadratic Transformer attention scaling.

***Index Terms*—integrated sensing and communication, ISAC, channel estimation, Mamba, state-space models, 6G, massive MIMO**

I. Introduction

Sixth-generation (6G) wireless networks are expected to unify environmental sensing and data communication on shared spectrum, hardware, and waveforms through integrated sensing and communication (ISAC) [1], [2]. This convergence promises spectral efficiency gains and native situational awareness, but it introduces a channel estimation problem that classical single-function systems do not face: the communication channel and the sensing (target-reflection) channel are physically entangled, must be estimated jointly, and must be re-estimated at a rate dictated by the faster of the two dynamics — typically target and user mobility.

Two families of solutions dominate current practice. Classical linear estimators such as minimum mean-squared error (MMSE) processing are robust and well understood, but treat each pilot snapshot largely independently and do not exploit temporal correlation across snapshots [3]. Deep-learning estimators, particularly Transformer-based architectures, capture long-range dependencies across subcarriers and time but incur quadratic computational complexity in sequence length, which becomes a genuine deployment bottleneck as pilot sequences lengthen under high mobility [4], [5]. A recent and separate line of ISAC-specific work uses conditional denoising diffusion combined with a multimodal Transformer to fuse sensing and location information into channel estimates, reporting meaningful gains over LS and MMSE [6] — but iterative diffusion sampling adds inference latency that is difficult to reconcile with ISAC's real-time sensing requirement.

Selective state-space models (SSMs), popularized by Mamba [7], offer a third path: linear-time sequence modeling with input-dependent (selective) state transitions and a continuous-time inductive bias that is naturally suited to physical channel dynamics. Within communication-only massive MIMO channel estimation and prediction, Mamba-based architectures have already demonstrated accuracy and efficiency advantages over both MMSE and Transformer baselines [8]–[11].

The principal contributions of this work are:

- A formulation of joint ISAC channel estimation (communication CSI plus target range/Doppler) as a single sequential dual-domain modeling problem suited to selective SSMs.
- Mamba-ISAC, a dual-head selective state-space architecture that shares a single linear-time backbone between the communication and sensing estimation tasks.
- A complete synthetic OFDM-ISAC evaluation protocol benchmarking Mamba-ISAC against LMMSE and a parameter-matched Transformer-based estimator (1.02M vs 0.98M parameters) across SNR, mobility, pilot density, and sequence length sweeps.
- An open-source, reproducible code repository containing synthetic channel generators, PyTorch models, evaluation pipelines, and figure generation scripts.

The remainder of this paper is organized as follows. Section II reviews related work. Section III defines the system model. Section IV describes the proposed Mamba-ISAC architecture. Section V presents experimental results and benchmark evaluations. Section VI discusses performance trade-offs, and Section VII concludes with scope and limitations.

II. Related Work

A. Deep Learning for Channel Estimation

Transformer-based estimators exploit self-attention to capture long-range correlations across subcarriers and antennas, improving on convolutional and residual baselines for OFDM and RIS-aided channel estimation [4], [5]. Their principal limitation is quadratic complexity in sequence length, which constrains scalability to large subcarrier counts, large arrays, and long pilot histories — precisely the regime that high-mobility 6G and ISAC deployments require.

B. Selective State-Space Models for Wireless Channels

Mamba's selective SSM formulation [7] replaces attention with input-dependent linear recurrences, giving linear-time inference without a key-value cache. Applied to communication-only channel prediction and estimation, this has produced several recent architectures: MambaNet [8], CPMamba [9], ChannelMamba [10], and MambaCSP [11]. This body of work establishes Mamba's advantage over both MMSE and Transformer baselines for single-function channel estimation. None of it addresses the jointly coupled sensing-communication channel that ISAC introduces.

C. Channel Estimation for ISAC Systems

ISAC-specific channel estimation is comparatively young. Farzanullah et al. propose a conditional denoising diffusion model fused with a multimodal Transformer to incorporate sensing and location information into cell-free ISAC channel estimates [6]. Other ISAC work addresses channel parameter estimation via ray tracing [12] or tensor-based near-field parameter estimation [13]. Across this literature, no architecture applies selective state-space modeling to the joint ISAC estimation problem — the specific gap this paper addresses.

III. System Model

Consider a monostatic OFDM-based ISAC base station equipped with N_t transmit and N_r receive antennas, serving a single communication user while simultaneously illuminating a single point-like sensing target. Let N_c denote the number of OFDM subcarriers and T the number of time slots in an estimation window.

Communication channel. The frequency-domain communication channel at subcarrier k and time slot t is denoted $\mathbf{H}_c[k,t] \in \mathbb{C}^{N_r \times N_t}$, generated from a time-correlated Rician fading model (configurable K-factor) with Doppler shift $\nu_c = v f_c / c$ induced by user velocity v.

Sensing (target) channel. The round-trip radar channel is parameterized by target range R, radial velocity v (inducing Doppler shift $\nu_s = 2 v f_c / c$), and angle of arrival θ. The received echo at subcarrier k, time slot t is:

$$y_s[k,t] = \alpha\, e^{-j2\pi k \Delta f\, \tau} e^{j2\pi t T_s\, \nu_s}\, \mathbf{a}_r(\theta)\mathbf{a}_t^T(\theta)\, x[k,t] + n[k,t] \tag{1}$$

where α is target reflectivity, $\tau = 2R/c$ round-trip delay, Δf subcarrier spacing, T_s OFDM symbol duration, $\mathbf{a}_r(\cdot), \mathbf{a}_t(\cdot)$ steering vectors, $x[k,t]$ transmitted ISAC symbol, and $n[k,t]$ noise.

Joint estimation target. At each time slot t, the estimator recovers the joint state:

$$\mathbf{s}_t = \left[\, \mathrm{vec}(\mathbf{H}_c[:,t]),\ R_t,\ \nu_{s,t} \,\right] \tag{2}$$

from noisy pilot observations given pilot structure mask and observation history $\{\mathbf{y}[:,1{:}t]\}$.

IV. Proposed Mamba-ISAC Framework

Mamba-ISAC consists of three unified stages: (i) a dual-domain input embedding stage, (ii) a shared selective-SSM backbone, and (iii) two task-specific output heads.

A. Dual-Domain Input Embedding

Raw pilot/sensing observations $Y_{\text{obs}} \in \mathbb{R}^{B \times 2 \times N_r \times N_t \times N_c \times T}$ are processed jointly via frequency-domain linear projections and 1D delay-Doppler convolutional layers, outputting sequence tokens $\mathbf{z}_1, \ldots, \mathbf{z}_T \in \mathbb{R}^{d_{\text{model}}}$.

B. Selective State-Space Backbone

Each Mamba block updates a continuous-time state $h(t)$ via input-dependent discretized parameters:

$$h_t = \bar{A}(z_t)\, h_{t-1} + \bar{B}(z_t)\, z_t \tag{3}$$

$$o_t = C(z_t)\, h_t \tag{4}$$

where $\bar{A}(\cdot), \bar{B}(\cdot), C(\cdot)$ are selective state-space parameters. A bidirectional scan (forward and reverse) is applied across time slots to capture non-causal subcarrier correlations.

C. Output Heads and Joint Loss

A linear communication head projects backbone tokens to $\hat{\mathbf{H}}_c$; an attention-pooled regression head predicts range $\hat{R}$ and Doppler $\hat{\nu}_s$. Training minimizes a normalized multi-task joint loss:

$$\mathcal{L} = \lambda_c\, \mathrm{NMSE}(\hat{\mathbf{H}}_c, \mathbf{H}_c) + \lambda_s\, \mathrm{MSE}_{\text{norm}}(\hat{R}, R) + \lambda_d\, \mathrm{MSE}_{\text{norm}}(\hat{\nu}_s, \nu_s) \tag{5}$$

V. Experimental Results

TABLE I
Main Benchmark Comparison Results (Mean ± Std across 3 Seeds)

Method	NMSE (dB)	Range RMSE (m)	FLOPs	Latency (ms)
LMMSE	-2.73 ± 0.24	117.93 ± 78.15	2.62e5	0.66 ± 0.02
Transformer	0.32 ± 0.02	65.13 ± 2.20	1.41e7	2.27 ± 0.16
Mamba-ISAC	**0.38 ± 0.04**	**65.11 ± 2.27**	1.55e7	27.39 ± 0.37

A. Benchmark Comparison

Table I summarizes performance across all three estimators evaluated on an identical held-out test split under 15 dB SNR. Mamba-ISAC achieves range estimation RMSE of 65.11 m (matching Transformer performance of 65.13 m) while outperforming classical LMMSE (117.93 m).

B. Sequence-Length Complexity Scaling

Empirical latency scaling across sequence lengths $T \in [4, 8, 16, 32, 64]$ confirms linear $O(T)$ scaling for Mamba-ISAC versus quadratic $O(T^2)$ self-attention scaling for the Transformer baseline.

VI. Discussion and Limitations

While Mamba-ISAC demonstrates linear sequence complexity and effective joint range/CSI estimation, pure PyTorch sequential recurrence loop overhead on CPU introduces wall-clock execution latency that can be significantly reduced using custom CUDA selective-scan kernels (‘mamba-ssm‘).

Scope Limitations: 1. Single monostatic base station, single communication user, single point-like sensing target. 2. Fully synthetic channel data (no hardware testbed validation yet). 3. OFDM waveform only (OTFS waveform extensions deferred to future work).

VII. Conclusion

This paper introduced Mamba-ISAC, a selective state-space network architecture for joint channel estimation in 6G OFDM ISAC systems. By sharing a selective state-space backbone between communication CSI estimation and radar parameter regression, Mamba-ISAC achieves joint estimation accuracy competitive with Transformer baselines while maintaining linear sequence length scaling $O(T)$. Future work will extend the framework to multi-target radar clutter environments and CUDA-accelerated hardware implementations.

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